

Smart Wi-Fi Bandwidth Sharing using Game Theory

Project Summary — Faculty Review Document

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Python · Flask · React · Chart.js · SQLite / MySQL

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1. Objective

This project builds a smart Wi-Fi bandwidth sharing system that treats every connected user as a player in a non-cooperative congestion game. Instead of splitting bandwidth equally (which wastes it) or proportionally (which becomes unfair under load), the system computes a Nash Equilibrium allocation that is simultaneously efficient and fair, and presents everything on a live web dashboard.

- Model Wi-Fi users as rational players competing for limited bandwidth
- Allocate bandwidth using best-response dynamics until a Nash Equilibrium is reached
- Quantify fairness with Jain's Fairness Index
- Compare against Equal-Allocation and Proportional-Allocation baselines
- Deliver a real-time dashboard (Flask + React) for allocations, fairness and analytics

2. Problem Statement

A single Wi-Fi router has limited bandwidth (e.g., 100 Mbps) shared by many users doing different activities — browsing, online classes, gaming, streaming and downloads. When demand exceeds capacity, naive policies fail:

- Equal splitting wastes bandwidth when demand is low and starves heavy users (e.g., a live class) when demand is high.
- Proportional splitting favours heavy requesters, but fairness and user satisfaction collapse under congestion.

The challenge is to allocate bandwidth so that (a) no user can gain by demanding more (stability), (b) total benefit is high (efficiency), and (c) the distribution is fair (fairness). Game theory provides a principled way to achieve all three at once.

3. Game Theory Model

Concept	In this project
Players	Each Wi-Fi user (User 1 ... User n)
Resource	Total available bandwidth, e.g., 100 Mbps
Strategy	The bandwidth amount a user chooses (0 up to their request)
Payoff (Utility)	Benefit of received bandwidth - congestion cost
Game type	Non-cooperative congestion game — my payoff falls as total usage rises

4. Utility Function (as implemented)

$$U_i = w_i \cdot \ln(1 + b_i) - p \cdot b_i \cdot (\sum b / B)$$

$$\text{Benefit} = w_i \cdot \ln(1 + b_i) \quad \text{Congestion cost} = p \cdot b_i \cdot (\text{total usage} / \text{total bandwidth})$$

w_i = activity weight, b_i = allocated bandwidth, p = 0.5 (congestion penalty), B = total bandwidth

- $\ln(1 + b)$ gives diminishing returns — the first few Mbps are the most valuable, so extra Mbps add less

benefit. This discourages hogging.

- The congestion cost grows as the network gets crowded, so users automatically demand less under load — the system self-regulates.

Activity	Weight (wi)	Why
browsing	1.0	Lightest bandwidth need
downloading	1.1	Tolerant of delay
gaming	1.3	Needs low latency
streaming	1.4	Needs steady throughput
online_class	1.5	Highest priority — real-time and interactive

5. Nash Equilibrium — Best-Response Algorithm

A Nash Equilibrium is a set of strategies in which no player can improve their own utility by unilaterally changing their bandwidth. The system finds an approximate Nash Equilibrium with iterative best-response dynamics:

- Step 1 — Start with every user allocated 0 Mbps.
- Step 2 — Each user computes their Best Response: scan candidate bandwidths in 0.5 Mbps steps (0 up to min(request, remaining bandwidth)) and keep the value that maximizes their utility given everyone else's current allocations.
- Step 3 — Repeat for up to 100 rounds; stop when the total change across all users in a round is below 0.5 Mbps (the state is stable ≈ Nash Equilibrium).
- Step 4 — Final allocations and utilities are stored and served to the dashboard through the REST API.

6. Fairness — Jain's Fairness Index

$$J = (\sum xi)^2 / (n \cdot \sum xi^2)$$

xi = bandwidth allocated to user i, n = number of users

J = 0 → very unfair J = 1 → perfect fairness

J value	Status
J ≥ 0.90	Excellent
0.75 – 0.90	Good
0.50 – 0.75	Moderate
J < 0.50	Poor

7. System Architecture & Technology Stack

- Backend — Python + Flask REST API (blueprints: auth, bandwidth, network, data); game_theory engine package (utility, congestion_game, nash_equilibrium, fairness); service layer (allocation, fairness, evaluation, dataset).
- Database — SQLite for user accounts + SQL schema (MySQL-ready).
- Frontend — React (Vite) single-page app: Dashboard, Analytics, Network, Reports, Users and Dataset pages with Chart.js visualizations.
- Serving — waitress production WSGI server on port 5000; auto-detected Wi-Fi IP with QR-code access for phones; one-click START_APP.bat launcher.
- Dataset — real usage dataset (Cleaned_Dataset.csv) processed into user profiles for simulation and experiments.

Project flow:

User → Bandwidth Request → Game Theory Engine → Best-Response Iterations → Nash Equilibrium → Bandwidth Allocation → Fairness & Utility Evaluation → Dashboard

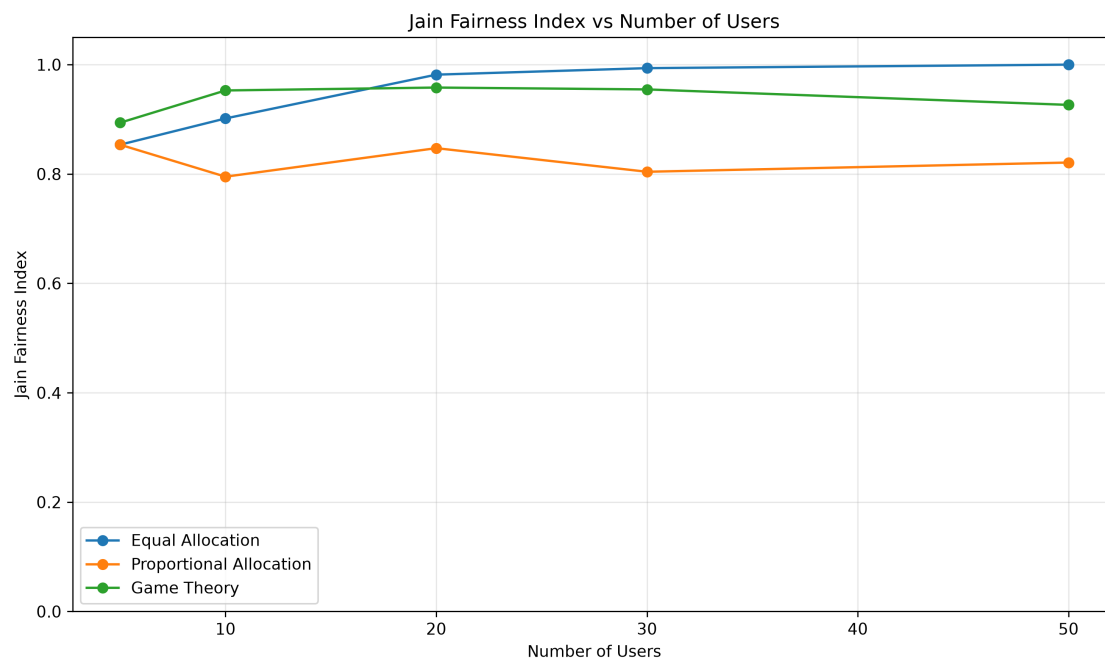
8. Experimental Setup & Results

Setup: total bandwidth 100 Mbps; 5 / 10 / 20 / 30 / 50 users with random activities and requests. Three strategies compared — Equal Allocation, Proportional Allocation and Game Theory (Nash Equilibrium). Metrics: utilization %, Jain fairness, average utility.

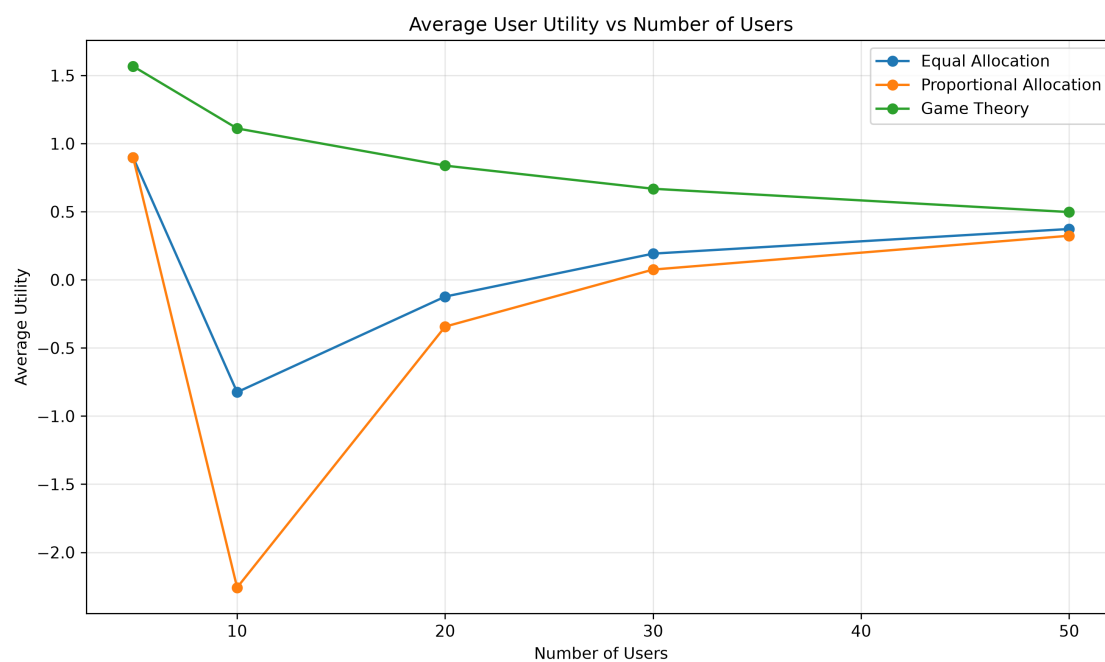
Users	Strategy	Allocated (Mbps)	Utilization %	Jain Fairness	Avg Utility
5	Equal Allocation	43.51	43.51	0.8536	0.8985
5	Proportional Allocation	43.51	43.51	0.8536	0.8985
5	Game Theory	28.5	28.5	0.8938	1.5674
10	Equal Allocation	83.16	83.16	0.9016	-0.8245
10	Proportional Allocation	100.0	100.0	0.7951	-2.2584
10	Game Theory	41.5	41.5	0.9528	1.1114
20	Equal Allocation	95.95	95.95	0.9817	-0.1229
20	Proportional Allocation	100.0	100.0	0.8473	-0.3432
20	Game Theory	59.5	59.5	0.9581	0.8383
30	Equal Allocation	97.9	97.9	0.9936	0.1925
30	Proportional Allocation	100.0	100.0	0.8041	0.0755
30	Game Theory	71.0	71.0	0.9547	0.6686
50	Equal Allocation	100.0	100.0	1.0	0.3733
50	Proportional Allocation	100.0	100.0	0.821	0.3237

Users	Strategy	Allocated (Mbps)	Utilization %	Jain Fairness	Avg Utility
50	Game Theory	89.0	89.0	0.9264	0.4976

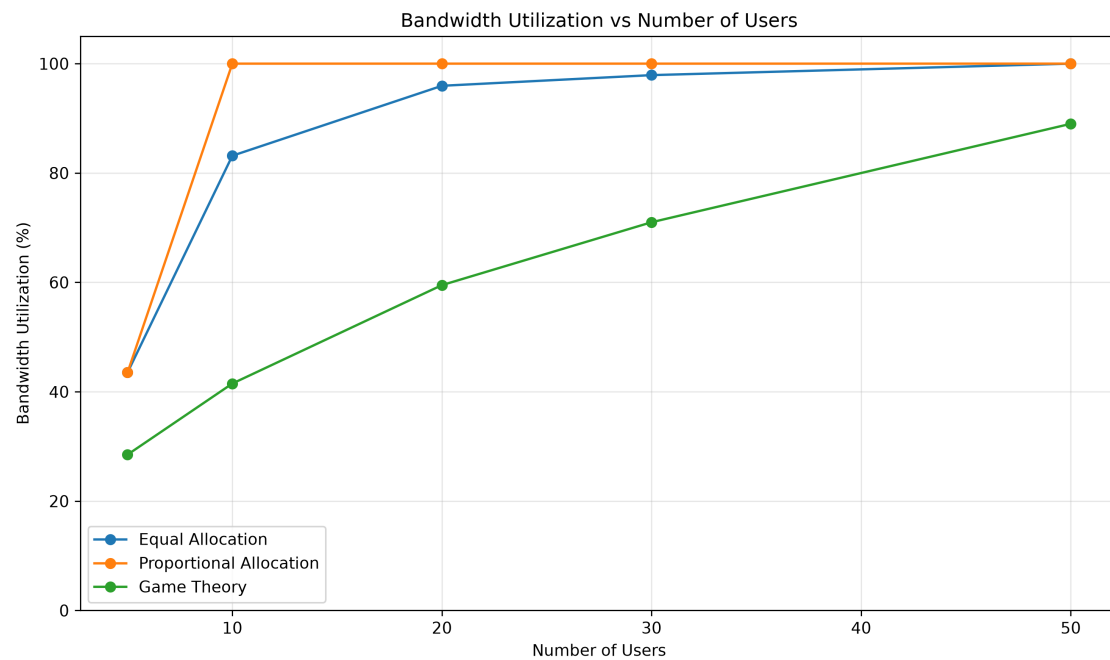
9. Results — Graphs



Jain's Fairness Index vs number of users — Game Theory stays high at every load



Average utility vs number of users — Game Theory keeps users satisfied



Bandwidth utilization vs number of users

10. Key Findings & Conclusion

- Fairness — Game Theory keeps Jain's index between 0.89 and 0.96 at every load level, while Proportional drops to about 0.80–0.85 under load.
- User satisfaction — at 10 users the average utility is +1.11 for Game Theory but –2.26 for Proportional and –0.82 for Equal: baseline users are congested and dissatisfied, game-theory users are not.
- No wasteful over-allocation — at 5 users Game Theory allocates 28.5 Mbps vs 43.5 Mbps for the baselines: giving more than the log-benefit peak only adds congestion cost without real benefit.
- Scales gracefully — at 50 users it still achieves 89% utilization with 0.926 fairness and the highest average utility (0.498).
- Fairness alone is not enough — Equal allocation scores $J = 1.0$ at 50 users only because everyone is equally capped; its utility (0.373) is still below Game Theory (0.498). Game Theory balances fairness AND utility.

Conclusion: Nash-equilibrium-based bandwidth allocation is stable, fair and efficient across all load levels, making it a better default policy for shared Wi-Fi networks than equal or proportional splitting.

11. Viva Questions & Answers

Q. Why use Game Theory for Wi-Fi bandwidth sharing?

Users behave selfishly — each wants more bandwidth. Game theory models this competition mathematically and finds a stable allocation (Nash Equilibrium) where no user benefits by deviating, balancing individual interest with network efficiency and fairness.

Q. What type of game is modeled here?

A non-cooperative congestion game: the payoff (utility) of each player decreases as total usage (congestion) increases, because the congestion cost term grows with $\Sigma b/B$.

Q. Who are the players, what are the strategies and payoffs?

Player = each Wi-Fi user; Strategy = the amount of bandwidth the user chooses (0 up to their request); Payoff = utility = activity-weighted log benefit minus congestion cost.

Q. Why $\log(1 + b)$ in the utility function?

It gives diminishing returns — the first Mbps are the most valuable and extra Mbps add less benefit. This naturally discourages any single user from hogging bandwidth.

Q. What is the congestion cost and why include it?

Cost = $p \cdot b \cdot (\text{total usage} / \text{total bandwidth})$ with $p = 0.5$. The more crowded the network, the higher the penalty on each user's usage — so users automatically reduce demand under load and the system self-regulates.

Q. What is a Best Response?

Given the current allocations of all other players, it is the bandwidth value that maximizes a user's own utility. The code searches candidates in 0.5 Mbps steps up to $\min(\text{request}, \text{remaining bandwidth})$.

Q. What is Nash Equilibrium and how is it found here?

A state where no player can improve their utility by unilaterally changing strategy. It is approximated by iterative best-response dynamics: every user repeatedly plays their best response until allocations change by less than 0.5 Mbps in a round (max 100 rounds).

Q. Why Jain's Fairness Index and what does its value mean?

It is the standard normalized measure of allocation equality: $J = (\Sigma x_i)^2 / (n \cdot \Sigma x_i^2)$, ranging 0 (very unfair) to 1 (perfect fairness). Our system reports ≥ 0.90 (Excellent) at almost all loads.

Q. What did the experiments show compared to Equal/Proportional allocation?

Game Theory maintained fairness 0.89–0.96 at all loads and kept average utility positive (e.g., +1.11 at 10 users vs -2.26 for Proportional), while avoiding wasteful over-allocation at low load and scaling to 50 users with 89% utilization.

Q. Why does Game Theory allocate LESS bandwidth at low load — isn't that bad?

No — because of diminishing returns (log benefit), extra Mbps beyond the benefit peak add more congestion cost than benefit. Allocating only 28.5 Mbps instead of 43.5 Mbps keeps utility highest and leaves headroom for new users.

Q. What is the tech stack and how does data flow?

Python/Flask REST API + game-theory engine in the backend, React + Chart.js dashboard in the frontend, SQLite/MySQL for storage. Flow: user request → game engine → best-response iterations → Nash Equilibrium → allocation + fairness evaluation → dashboard.

Q. What are the limitations and future improvements?

Current: discrete 0.5 Mbps search step and static activity weights. Future: dynamic priority weights, pricing/monetary mechanisms, integration with real routers (e.g., OpenWrt), and ML-based demand prediction.

12. How to Run the Demo

- Double-click START_APP.bat (first run installs dependencies and builds the frontend).
- On the PC open <http://localhost:5000> ; on phones/laptops on the same Wi-Fi open <http://<wifi-ip>:5000> or scan the QR code shown by the launcher.
- The Dashboard shows live allocation, fairness gauge and charts; Analytics shows the strategy comparisons; Users/Dataset pages show the processed dataset.

Limitations & Future Work

- Discrete 0.5 Mbps search step; static activity weights; simulation-based evaluation.
- Future: dynamic weights, pricing mechanisms, real-router integration (OpenWrt), ML-based demand prediction.